

Radio Map Assisted Path Planning for UAV Anti-Jamming Communications

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Abstract—The communications quality of unmanned aerial vehicle (UAV) has been brought into focus increasingly, which can be severely affected by other users, as well as complex topography. In this letter, we propose a path planning algorithm for UAV anti-jamming communications with the joint assistance of two kinds of radio maps—signal power map and signal to interference plus noise ratio (SINR) map, which contain the abundant information of path loss caused by the topography and interference from jammers. We show how to generate these maps from real topographic data, and provide the optimal UAV path planning by adopting the power and the SINR maps to the heuristic and graphic based A* algorithm. Simulations with various settings for areas with complex topography in real world, show that the proposed algorithm can considerably improve the efficiency by minimizing the UAV flying distance for anti-jamming communications.

Index Terms—UAV, path planning, anti-jamming communication, radio map.

I. INTRODUCTION

UNMANNED Aerial Vehicle (UAV) can be applied to various flying missions, including disaster relief, electronic interference, and surveillance scenarios. In many UAV missions, the quality of the UAV to base station link is crucial, which can be effected by electromagnetic attenuation and co-channel interference [1]. Not only to meet the requirement of link quality, but also for the flying efficiency, communication-connected UAV path planning has been widely studied [2]–[8]. In [2], the authors studied the optimization of UAV flight paths when there were interference signals in the UAV relay communications system. The UAV flight path is optimized to maximize the signal-to-interference-plus-noise ratio (SINR). In [3], the authors proposed a jamming detection-localization-evasion framework for estimating the location of disturbances and interference signal power. In [4], the authors investigated anti-jamming trajectory design for UAV-enabled wireless sensor networks with practical probabilistic line-of-sight (LOS) model. In [5], the authors studied 3D path planning for UAV based on radio map in cellular

communication scenario, and the optimal/suboptimal solutions were given with the SINR map. In [6], the authors studied the UAV-enabled radio access network, in which the UAV periodic flight duration or mission completion time were minimized, while satisfying the target data requirement of each ground user. In [7], the authors considered the problem of online UAV path planning for joint detection and tracking of multiple unknown radio-tagged objects. In [8], the authors studied a UAV enabled two-user interference channel. Many of the above works require, radio maps, the radio propagation environment simulation and reconstruction [9], to evaluate link quality comprehensively. While, this communication-connected UAV has many practical applications such as wireless data collection [4], cargo delivery [5], target detection and tracking [7].

However, there are still two major concerns for the existing works on UAV path planning for anti-jamming communications. The first is that the radio map employed for path planning is not generated from real topographic data, and this affects the precision of the radio map employed for path planning. The second is that, most existing works only employ SINR map. However, in practice, the received power is also critical important, especially for low cost receiver which is less sensitive to received signal. To our best knowledge, there has been no work on communication-connected UAV path planning with the assistance of both SINR map and power map generated by real world topographic data. In this paper, We show how to generate received signal power map and the SINR map from real topographic data, based on which, we propose an A* based algorithm for path planning of UAV anti-jamming communications to improve the flight efficiency of the communication-connected UAV.

II. RADIO MAP ASSISTED UAV PATH PLANNING

A. Problem Formulation

Fig. 1 illustrates the UAV path planning problem with one base station and N jammers in an area of complex topography. The UAV needs to fly from the start coordinate to the end coordinate while communicating with the base station. The jammers, as well as the highly complex topography affect the link quality of the UAV. We want to find the shortest path from the start to the end while the wireless link qualities of the UAV are beyond given thresholds.

We define the UAV flying region $\mathbf{X}$ as a rectangle, which is divided into $A \times B$ grids, where A and B are the numbers of divisions for the longitude and the latitude, respectively, and the length and width of each grid are both δ . Let $x_{i,j} \in$

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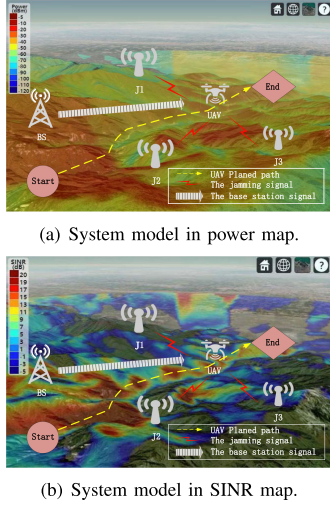


Fig. 1. Illustration of path planning for UAV anti-jamming communications in radio maps.

$\{(-90^\circ, 90^\circ), (-180^\circ, 180^\circ)\}$, in latitude and longitude, be the center coordinate of the (i, j) grid of $\mathbf{X}$. Let the $A \times B$ matrices $\mathbf{M}^{(P)}$ and $\mathbf{M}^{(S)}$, denote the power map and the SINR map, respectively. The (i, j) element of $\mathbf{M}^{(P)}$ is the received power of the UAV at the coordinate $x_{i,j}$, and that of $\mathbf{M}^{(S)}$ is SINR of UAV at the coordinate $x_{i,j}$. In Section II-B, we will show how to generate $\mathbf{M}^{(P)}$ and $\mathbf{M}^{(S)}$ from real world topographic data. We further define $\{z_1, \dots, z_K\}$ as the flying path of the UAV, where $z_k = x_{i(z_k), j(z_k)}$ is an element of $\mathbf{X}$ for all $k \in \{1, \dots, K\}$. The discrete UAV path planning with escaping multiple jammers can be formulated as

$$\min_{K, \{z_1, \dots, z_K\}} \sum_{k=1}^K \|z_{k+1} - z_k\| \quad (1)$$

$$\text{s.t.} \begin{cases} p(z_k) \geq p_{\min}, s(z_k) \geq s_{\min} \\ p(z_k) = m_{i(z_k), j(z_k)}^{(P)}, s(z_k) = m_{i(z_k), j(z_k)}^{(S)} \\ \|z_{k+1} - z_k\| \in \{0, \delta, \sqrt{2}\delta\} \end{cases} \quad (2)$$

where $p_{\min}$ and $s_{\min}$ are the received signal power and SINR thresholds. Note that $\|z_{k+1} - z_k\| \in \{0, \delta, \sqrt{2}\delta\}$ means for each time, the UAV can stay at the current grid, move to the east/west/north/south grid, or move to the north-west/northeast/southwest/southeast grid.

B. Radio Maps Generation From Real World Topographic Data

Note that $m_{i,j}^{(P)}$ can be evaluated by Friis path loss (in dB) equation [1], [10]:

$$m_{i,j}^{(P)} = P_t + G_t + G_m - \omega(\text{Tgra}, l_{BS}) \quad (3)$$

where G_t is the transmitter gain, G_m is the receiver gain, and $\omega(\cdot)$, depending on the topographic data Tgra and the base station position l_{BS} , are the path loss from the transmitter to the receiver. Depends on the propagation scenario, one can pick the most suitable propagation model for $\omega(\cdot)$. In this paper, while we pick up we employ the Longley-Rice model [11], which is widely

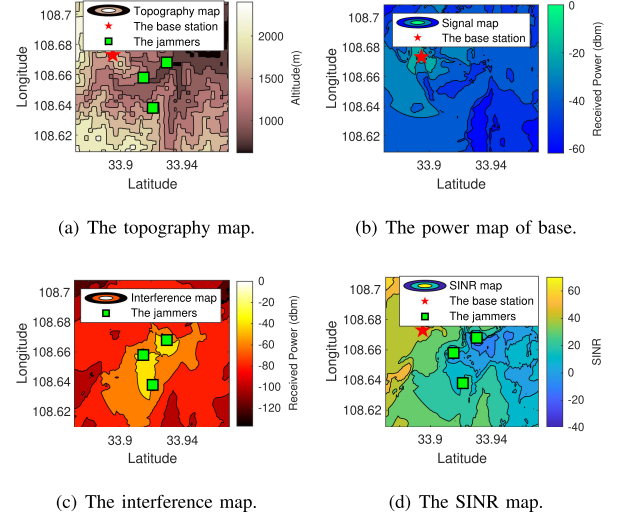


Fig. 2. The topographic, the power, the interference, and the SINR maps for an area in Taiping National Forest Park, Shaanxi Province, China.

employed for irregular and complex terrain for the path loss from the UAV to the ground [12]:

$$\omega(\text{Tgra}, l_{BS}) = \omega_{free} + \begin{cases} \max(d_{\min}, \omega_{rl} + n_1 d + n_2 \lg d), & d_{\min} \leq d \leq d_{rl} \\ \omega_{dl} + c_d d, & d_{rl} \leq d \leq d_{dl} \\ \omega_{sl} + c_s d, & d_{dl} \leq d \end{cases} \quad (4)$$

where ω_{free} is the free space propagation path loss, d is the path distance from UAV to the base station, d_{rl} is the ground plane distance, which is the boundary point of LOS propagation and diffraction propagation, and d_{dl} is the boundary point where the diffraction loss is equal to the scattering loss, ω_{rl} , ω_{dl} and ω_{sl} are the gains of the LOS, the diffraction, and scattering, respectively, c_d and c_s are the coefficients of diffraction and scattering, respectively, n_1 and n_2 are the coefficients of the propagation path loss. The above parameters depend on Tgra, which can be obtained from the Global Multi-resolution Terrain Elevation Data [13], and by using the Antenna Toolbox in Matlab [14], we can obtain $m_{i,j}^{(P)}$. For the SINR map $\mathbf{M}^{(S)}$,

$$m_{i,j}^{(S)} = \frac{m_{i,j}^{(P)}}{\sum_{n=1}^N m_{i,j}^{(J_n)} + \sigma^2}; \quad (5)$$

where $m_{i,j}^{(J_n)}$ is the interference power at $x_{i,j}$ with the n -th jammer position being l_{J_n} , and can be obtained in the way as $m_{i,j}^{(S)}$, σ^2 is the noise power at the UAV receiver. Examples of the power, interference, and SINR maps generated from real topographic data are shown in Fig. 2.

C. The Proposed Algorithm for UAV Path Planning

In this section, we propose an efficient heuristic and graphic based algorithm to solve the optimization problem (1) efficiently. Based on the power and SINR maps, we can obtain a map of

Algorithm 1: The Proposed Algorithm Based on SINR and Power Maps.

Input: Mission area $[\mathbf{X}]_{A \times B}$, feasible-zone map $[\mathbf{Y}]_{A \times B}$, start and end coordinates z_1 and z_F

- 1 Iteration number $t = 1$;
- 2 Open list $OPEN = \{z_1\}$, closed list $CLOSED = \emptyset$;
- 3 Evolution matrix $[\mathbf{F}]_{A \times B} = [\mathbf{Inf}]_{A \times B}$, // Elements initialized to ∞
- 4 $[\mathbf{G}]_{A \times B} = [\mathbf{Inf}]_{A \times B}$;
- 5 **while** $z_F \in CLOSED$ **or** $OPEN \neq \emptyset$ **do**
- 6 $X_{adj}(z_t) = \{x_{p,q} : \|x_{p,q} - z_t\| \leq \sqrt{2}\delta, y_{p,q} = 1, x_{p,q} \notin CLOSED\}$;
- 7 **for each** $x_{p,q} \in X_{adj}(z_t)$ **do**
- 8 **if** $g_t(x_{p,q}, \mathbf{Y}) < [\mathbf{G}]_{p,q}$ **then**
- 9 $[\mathbf{G}]_{p,q} = g_t(x_{p,q}, \mathbf{Y})$;
- 10 $[\mathbf{F}]_{p,q} = f_t(x_{p,q}, \mathbf{Y})$;
- 11 Update z_t as the *parent coordinate* of $x_{p,q}$;
- 12 **end**
- 13 **end**
- 14 $OPEN = OPEN \cup X_{adj}(z_t)$;
- 15 Find $x_{p,q} \in OPEN$ satisfying $\min \{f_t(x_{p,q}) : x_{p,q} \in OPEN\}$;
- 16 Drop z_t from $OPEN$ and add z_t to $CLOSED$;
- 17 $t = t + 1$;
- 18 $z_t = x_{p,q}$;
- 19 **end**
- 20 **if** $z_F \in CLOSED$ **then**
- 21 Planned path $Z = \{z_1, z_2, \dots, z_F\}$ // Optimal
- 22 **else**
- 23 Planned path $Z = \emptyset$; // Infeasible;
- 24 **end**

feasible zone $\mathbf{Y}$, an $A \times B$ matrix, with elements:

$$y_{i,j} = \begin{cases} 1, & m_{i,j}^P \geq p_{min} \text{ and } m_{i,j}^S \geq s_{min} \\ 0, & m_{i,j}^P < p_{min} \text{ or } m_{i,j}^S < s_{min}. \end{cases} \quad (6)$$

Note that $y_{i,j} = 1$ means both the received signal power and SINR are beyond the thresholds at $x_{i,j}$.

A* algorithm is a well known path planning algorithm based on both graphic theory and heuristic strategy [15]. It guarantees the optimal solution with the advantage searching efficiency, compared with other graph based and machine learning based algorithms.

In the A* algorithm, for each iteration, the evaluation function for the current coordinate $x_{i,j}$

$$f_t(x_{i,j}, \mathbf{Y}) = g_t(x_{i,j}, \mathbf{Y}) + h(x_{i,j}), \quad (7)$$

is to estimate the length of the shortest path from z_1 to z_F passing $x_{i,j}$, where

$$g_t(x_{i,j}, \mathbf{Y}) = \|x_{i,j} - z_1\|_d = \sum_{k=1}^{\kappa} \|z_{k+1} - z_k\| \quad (8)$$

is the real length of the shortest path from z_1 to $x_{i,j}$ in $\mathbf{Y}$, and

$$h(x_{i,j}) = \|x_{i,j} - z_F\|_{es}, \quad (9)$$

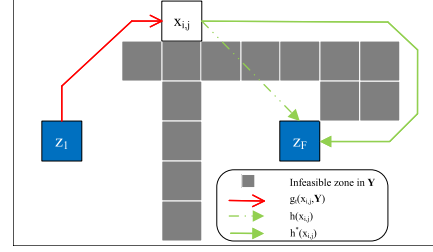


Fig. 3. Illustration of the evolution function of A* algorithm.

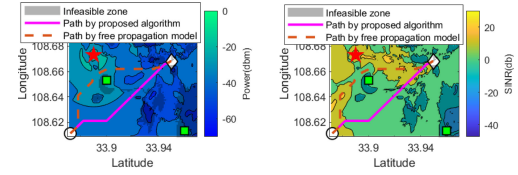
is the estimation of the length of the path from $x_{i,j}$ to z_F . Let $h^*(x_{i,j})$ denote the real distance from $x_{i,j}$ to z_F , it was proved that whatever the cost function $\|x_{i,j} - z_F\|_{es}$ is, as long as $h(x_{i,j}) \leq h^*(x_{i,j})$ for all $x_{i,j}$, the A* algorithm converges to the optimal solution [15]. Fig. 3 shows an example for the evaluation function $f_t(x_{i,j}, \mathbf{Y})$, where $h(x_{i,j})$ is chosen to be Euclidean distance, which satisfies $h(x_{i,j}) \leq h^*(x_{i,j})$ for all $x_{i,j}$. Based on the evaluation function (7) and the map of feasible zone $\mathbf{Y}$, we can find the optimal solution for (1), via the A* star algorithm, which is summarized in Algorithm 1.

III. SIMULATIONS

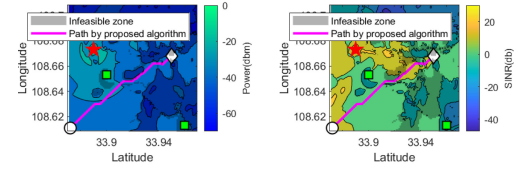
In this section, we provide simulations based on real topographical data from Taiping National Forest Park, shown in Fig. 2(a), a mountainous area in Xi'an, Shaanxi, China, where the wireless propagation is highly complex. The size of the area $11.1 \text{ km} \times 11.1 \text{ km}$. Both the power maps and the SINR maps are generated based on the longley-rice propagation model described in Section II-B. We consider the case that there are one base station and three jammers, and assume that the transmission powers and positions of base station and jammers are known to the UAV.

Fig. 4 shows the simulations for the scenario that the transmission power of the base station is 10 W, and those of the jammers are all 1 W. Four kinds of SINR and received power thresholds are considered. We can see that for the case $s_{min} = 0 \text{ dB}$ and $p_{min} = -70 \text{ dBm}$, both the algorithms can find a path from the start coordinate to the end coordinate, and the path found by the proposed algorithm is considerable shorter than that found by the algorithm based on free space propagation. When $s_{min} = 0$ and $p_{min} = -40 \text{ dBm}$, which corresponds to a less sensitive receiver, the proposed algorithm can still find a path, while the algorithm based on free space propagation fails. For the case $s_{min} = 3 \text{ dB}$ and $p_{min} = -70 \text{ dBm}$, both the algorithms can find a path from the starting coordinate to the end coordinate, and the path found by the proposed algorithm is a bit shorter than that found by the algorithm based on free space propagation. When $s_{min} = 0$ and $p_{min} = -40 \text{ dBm}$, which corresponds to a less sensitive receiver, the proposed algorithm can still find a path, and the algorithm based free space propagation fails again.

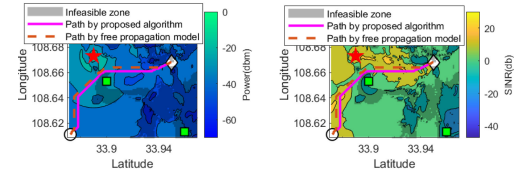
In Fig. 5 the transmission powers of the base station and the jammers are all 1 W, i.e., a worse wireless condition than that of Fig. 4. Still, four kinds of SINR and received power thresholds are considered. Note that the power thresholds are set to be different from those in Fig. 4, as the transmission powers from the base stations are different. For all the thresholds settings the



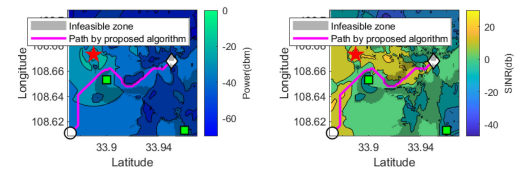
(a) Paths plotted in power map: $s_{\min} = 0$ dB, $p_{\min} = -70$ dBm. (b) Paths plotted in SINR map: $s_{\min} = 0$ dB, $p_{\min} = -70$ dBm.



(c) Path plotted in power map: $s_{\min} = 0$ dB, $p_{\min} = -40$ dBm. (d) Path plotted in SINR map: $s_{\min} = 0$ dB, $p_{\min} = -40$ dBm.



(e) Paths plotted in power map: $s_{\min} = 3$ dB, $p_{\min} = -70$ dBm. (f) Paths plotted in SINR map: $s_{\min} = 3$ dB, $p_{\min} = -70$ dBm.

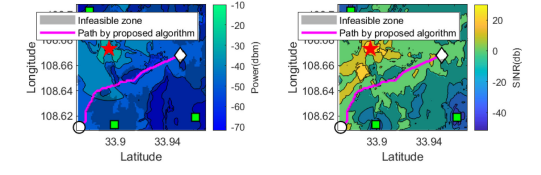


(g) Paths plotted in power map: $s_{\min} = 3$ dB, $p_{\min} = -40$ dBm. (h) Paths plotted in SINR map: $s_{\min} = 3$ dB, $p_{\min} = -40$ dBm.

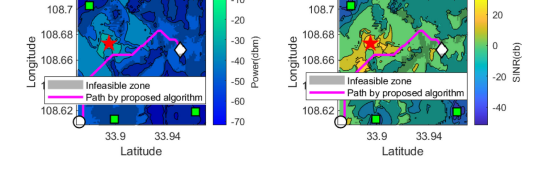
Fig. 4. Illustrations of the UAV paths found in the scenario that the transmission powers of the base station and jammers are 10 W and 1 W, respectively. In some cases the free space propagation based algorithm fails.

TABLE I
SUMMARIZE OF SIMULATION RESULTS FOR THE PATH PLANNING ALGORITHMS
IN VARIOUS SCENARIOS AND SETTINGS

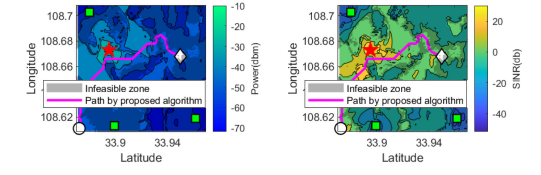
Path planning strategy	Communication Scenario			Applicability	Flying distance (Km)
	Scenario	SINR threshold	Power threshold		
A* algorithm with radio map (Proposed Algorithm)	Scenario 1	0db	-70dbm	✓	11.278
		0db	-40dbm	✓	11.370
		3db	-70dbm	✓	12.839
		3db	-40dbm	✓	14.348
	Scenario 2	0db	-70dbm	✓	12.254
		0db	-50dbm	✓	14.673
		3db	-70dbm	✓	15.886
		3db	-50dbm	✓	15.886
A* algorithm with free propagation model	Scenario 1	0db	-70dbm	✓	12.969
		0db	-40dbm	×	—
		3db	-70dbm	✓	13.099
		3db	-40dbm	×	—
	Scenario 2	0db	-70dbm	×	—
		0db	-50dbm	×	—
		3db	-70dbm	×	—
		3db	-50dbm	×	—



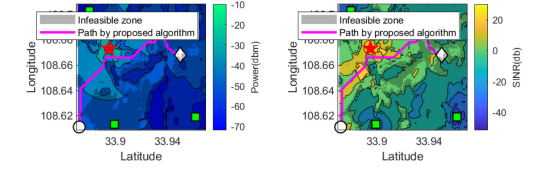
(a) Path plotted in power map: $s_{\min} = 0$ dB, $p_{\min} = -70$ dBm. (b) Path plotted in SINR map: $s_{\min} = 0$ dB, $p_{\min} = -70$ dBm.



(c) Path plotted in power map: $s_{\min} = 0$ dB, $p_{\min} = -50$ dBm. (d) Path plotted in SINR map: $s_{\min} = 0$ dB, $p_{\min} = -50$ dBm.



(e) Path plotted in power map: $s_{\min} = 3$ dB, $p_{\min} = -70$ dBm. (f) Path plotted in SINR map: $s_{\min} = 3$ dB, $p_{\min} = -70$ dBm.



(g) Path plotted in power map: $s_{\min} = 3$ dB, $p_{\min} = -50$ dBm. (h) Path plotted in SINR map: $s_{\min} = 3$ dB, $p_{\min} = -50$ dBm.

Fig. 5. Illustrations of the UAV paths found in the scenario that the transmission powers of the base station and jammers are all 1 W. Note that the free space propagation based algorithm fails in this scenario.

algorithm based on free space propagation fails to find a path, while the proposed algorithm successfully find a path for all cases. The simulation results are summarized in Table I. As we can see that with the assistance of radio maps, a considerably better path planning can be obtained, particularly in bad RF propagation condition, in which the free space propagation based algorithm may fail.

IV. CONCLUSION

In this letter, we proposed a radio map assisted path planning algorithm for UAV anti-jamming communications. We show how to generate both the SINR map and power map from real world topographic data, and proposed a UAV path planning algorithm by adopting the radio maps to the A* algorithm. Simulation results with real world topographic data under various settings showed that the proposed algorithm can considerably outperform the algorithm based on free space propagation, particularly in complex and bad electromagnetic propagation environment. Future work could be setting up real experiment to further verify the proposed algorithm.

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